Learning Spectral Transition Dynamics from Fish Schools for Bio-Inspired Swarm Control

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1 Background

Collective motion in fish schools is a prominent example of biological self-organization, where coherent group behavior emerges from local interactions among individuals rather than from centralized control. Fish schools can display polarized motion, milling, collective turning, expansion, compression, fragmentation, and regrouping. These behaviors arise through sensory feedback, neighbor interactions, hydrodynamic effects, and environmental constraints, producing group-level dynamics that are richer than the behavior of any individual fish [1, 2, 3].

Classical models of collective motion, such as attraction–alignment–repulsion models, Reynolds-style flocking models, and self-propelled particle models, have shown how simple local rules can generate coordinated group behavior [4, 5, 1]. However, these models often focus on reproducing stable behavioral states. They are less suited to capturing how a school moves between states, especially when the group does not settle into one regime but alternates between several competing collective configurations.

A central distinction in this project is between stationary phases and multistable phases. In a stationary phase, the school maintains a relatively stable collective organization, such as persistent polarized swimming or stable milling. In a multistable phase, several collective modes coexist, and the school may switch between them over time. Such switching behavior is consistent with the broader view of collective animal motion as a nonlinear dynamical process shaped by feedback, perturbation, and changing interaction networks [6, 7, 8].

Spectral methods provide a natural framework for analyzing such dynamics. Graph Laplacian spectra can describe the changing interaction topology of the school, while Dynamic Mode Decomposition and Koopman-inspired observables can reveal dominant spatiotemporal modes, frequencies, and stability patterns [9, 10, 11, 12]. These tools allow the school to be represented not only as individual trajectories, but as a low-dimensional dynamical structure whose modes evolve over time.

This representation makes it possible to study transitions directly. Instead of asking only whether a school is schooling, milling, expanding, or regrouping at a given moment, the project asks how the school moves between these regimes. Changes in spectral energy, graph connectivity, eigenvalue structure, modal dominance, and spectral entropy may indicate when a stationary phase is becoming unstable or when a multistable phase is emerging. Similar ideas have been used in dynamical-systems analysis to identify coherent structures, regime shifts, and low-dimensional coordinates of complex nonlinear systems [13, 14, 15, 16].

The same spectral transition representation can also support artificial swarm control. Multi-agent reinforcement learning can use the learned spectral space as an objective, training simulated and robotic swarms to reproduce not only fish-like collective states but also the transition geometry between them. This creates a bridge between biological collective behavior and bio-inspired robotics, where the goal is not simple imitation of trajectories, but transfer of the underlying dynamical logic of collective phase transitions into artificial multi-agent systems [17, 18, 19, 20].

2 Previous Work

Research on collective animal motion has established fish schools as a central model for studying distributed coordination, self-organization, and group-level decision-making. Early computational models showed that coherent collective motion can emerge from simple local interaction rules, typically based on attraction, alignment, and repulsion among neighboring individuals. Reynolds' boids model demonstrated this principle in artificial flocking systems, while the Vicsek model formalized phase transitions in self-propelled particle systems, showing how ordered motion can emerge from local alignment under noise [4, 5]. Couzin et al. later extended this perspective to biological groups, showing how local interaction zones can generate collective memory, spatial sorting, and coordinated group movement [1].

Empirical studies of fish schools have refined these models by estimating interaction rules directly from observed trajectories. Katz et al. inferred the structure and dynamics of interactions in schooling fish and showed that individual responses depend strongly on neighbor position and group context, providing evidence that fish-school coordination is shaped by structured social interactions rather than by uniform metric averaging [2]. Herbert-Read et al. similarly investigated shoaling fish and demonstrated that interaction rules can be inferred from movement data, linking individual response functions to group-level coordination [7]. Lopez et al. reviewed the progression from behavioral analysis to mathematical models of fish schooling, emphasizing that collective motion models must account for both individual-level mechanisms and emergent group-level structure [3].

Beyond local-rule models, previous work has shown that fish schools can exhibit multiple collective regimes, including polarization, milling, swarming, and transitional behavior. Tunstrøm et al. studied collective states and multistability in fish schools, showing that groups may switch between distinct macroscopic states rather than remaining in a single stable regime [21]. This work is especially relevant to the present proposal because it frames fish schooling as a dynamical system with multiple possible collective attractors. However, much of the existing literature still treats these regimes primarily as behavioral classes. The proposed PhD extends this direction by placing the emphasis on the transition structure between stationary and multistable phases, rather than on the classification of states alone.

Graph-based approaches have also contributed significantly to the analysis of collective behavior. In biological groups, interaction topology can be more informative than metric distance alone. Ballerini et al. showed in starling flocks that interaction structure may depend on a fixed number of topological neighbors rather than on all individuals within a fixed metric radius [6]. This insight is highly relevant for fish schools, where local neighborhoods change continuously as individuals move, occlude one another, and reorganize. Graph Laplacian methods provide a mathematical framework for representing such dynamic interaction networks, where eigenvalues and eigenvectors encode connectivity, cohesion, fragmentation, and smooth variation over the group [9]. These tools support the proposed use of graph spectra to detect when a stationary collective phase begins to destabilize or when competing modes appear during multistable behavior.

Spectral and operator-theoretic methods have become increasingly important for analyzing complex dynamical systems. Dynamic Mode Decomposition provides a data-driven method for extracting coherent spatiotemporal modes, modal amplitudes, and characteristic frequencies from time-series data [10, 11]. Koopman operator theory extends this perspective by representing nonlinear dynamics through the evolution of observables in a lifted space, making it possible to identify coordinates in which nonlinear systems exhibit approximately linear structure [13, 14, 12]. These methods have been widely used in fluid dynamics and nonlinear systems analysis, but their application to fish-school phase transitions remains comparatively underdeveloped. This gap motivates the proposed use of DMD, Koopman-inspired observables, and graph spectra to model collective switching as a dynamical object in its own right.

Machine learning has also been applied to collective motion, particularly for representation learning, behavioral classification, and interaction inference. Recent neural approaches have attempted to recover collective interaction rules from data, including attention-based models that infer which neighbors influence individual motion [22]. Such methods are valuable because they can capture nonlinear dependencies that are difficult to specify manually. However, many data-driven approaches still focus on predicting individual trajectories or classifying behavioral states. The present proposal instead aims to learn a latent spectral transition space, in which collective states are represented by spectral signatures and behavioral switching is represented as movement through this space. This shifts the learning target from individual-level prediction to population-level dynamical geometry.

In swarm robotics, biological collective motion has long inspired decentralized control algorithms. Classical swarm systems often use local rules similar to attraction, alignment, and repulsion, while more recent approaches use learning-based methods to discover policies for coordination. Rubenstein et al. demonstrated large-scale programmable self-assembly in a thousand-robot swarm, showing the feasibility of robust collective behavior in physical robotic systems [17]. In reinforcement learning, Lowe et al. introduced multi-agent actor–critic methods for mixed cooperative–competitive environments, and Hüttenrauch et al. proposed deep reinforcement learning representations specifically designed for swarm systems, emphasizing permutation invariance and scalability to varying numbers of agents [18, 19]. These works provide the technical foundation for using MARL to train artificial swarms, but they do not directly address how to reproduce biological phase-transition structure.

The proposed research therefore builds on, but differs from, three major lines of previous work. First, it builds on biological and computational models of fish schooling, but shifts attention from stable collective states to transitions between stationary and multistable phases. Second, it builds on spectral and Koopman-based analysis of dynamical systems, but applies these tools to the latent phase structure of collective animal motion. Third, it builds on swarm robotics and MARL, but uses spectral transition geometry as the learning objective rather than relying only on hand-crafted task rewards, direct trajectory imitation, or static behavioral targets. The intended contribution is a unified framework in which fish-school dynamics are analyzed, learned, and transferred as a structured landscape of collective modes and regime transitions.

3 Research Objectives

The objective of this PhD is to develop a spectral, transition-centered framework for analyzing and reproducing collective fish-school dynamics, with emphasis on transitions between stationary and multistable phases. Rather than treating schooling, milling, expansion, and regrouping as isolated behavioral states, the project will model them as regimes within a shared dynamical landscape.

The first objective is to construct a compact spectral representation of fish-school motion. Individual trajectories will be transformed into population-level observables, including graph Laplacian spectra, Dynamic Mode Decomposition modes, and Koopman-inspired features. This representation will capture collective structure while remaining robust to changing neighbors, variable group size, and imperfect tracking.

The second objective is to characterize stationary and multistable collective phases. Stationary phases will be defined as regimes with relatively stable macroscopic organization, such as persistent polarization or milling. Multistable phases will be defined as regimes in which several collective modes coexist, compete, or alternate over time. These phases will be quantified through modal energy distributions, spectral entropy, graph connectivity, dwell times, recurrence patterns, and transition probabilities.

The third objective is to analyze transitions between phases as primary dynamical events. The project will identify spectral markers that precede or accompany behavioral switching, such as eigenvalue drift, redistribution of spectral energy, weakening of graph connectivity, loss of dominant modes, and emergence of competing modes. This will allow the project to study not only where the school is behaviorally, but how close it is to reorganizing.

The fourth objective is to construct a latent spectral transition space. In this space, each time window of collective motion will be represented by its spectral signature, and behavioral evolution will appear as a trajectory through this space. Stationary regimes are expected to form stable regions, while multistable dynamics are expected to appear as movement among competing regions or attractor-like structures.

The fifth objective is to use this spectral transition space as an objective for multi-agent reinforcement learning. Artificial swarms will be trained not merely to imitate fish trajectories, but to reproduce the collective modes and transition geometry of natural schools while satisfying constraints such as collision avoidance, cohesion, energy efficiency, and task performance.

The final objective is to test whether the learned transition dynamics can transfer from biological fish schools to simulated agents and physical robotic swarms. The key evaluation will be whether dominant spectral modes and biologically plausible switching patterns persist across different embodiments, despite noise, sensing limitations, and actuation constraints.

4 Contribution

This research will contribute to the study of collective behavior by reframing fish schooling as a phase-structured dynamical phenomenon rather than as a set of discrete behavioral categories. By emphasizing transitions between stationary and multistable phases, the work will help shift the field from state classification toward a deeper understanding of how collective systems destabilize, reorganize, and adapt over time. This perspective can provide new quantitative language for studying flexibility, responsiveness, and group-level decision-making in animal collectives, particularly in contexts such as escape, regrouping, exploration, compression, and collective reorientation.

Methodologically, the research will contribute an integrated framework that connects spectral graph theory, dynamical-systems analysis, and data-driven modeling for the study of biological collectives. For swarm robotics, it will offer a biologically grounded control perspective in which artificial swarms are not designed only to reproduce fixed formations or isolated tasks, but to preserve adaptive switching logic observed in natural systems. More broadly, this work will contribute a transferable conceptual framework for understanding complex multi-agent systems across biological, computational, and embodied domains, showing that the organizing principles of collective motion may lie not only in the states a group occupies, but in the structured pathways by which it moves between them.

5 Methodology

The methodology is based on representing fish-school behavior through three complementary observable families: graph Laplacian spectra, Dynamic Mode Decomposition, and Koopman-inspired observables. These observables will be used to analyze collective motion as a dynamical system whose behavior evolves through stationary phases, multistable phases, and transitions between them. The project will therefore avoid relying on broad hand-crafted behavioral descriptors as final analytical variables; instead, such quantities may be used only as preprocessing inputs when needed to construct the three main observable spaces (Fig. 1).

5.1 Data Acquisition and State Reconstruction

Fish-school dynamics will be analyzed from video recordings collected under controlled experimental conditions or from existing datasets. The recordings will include spontaneous swimming as well as perturbation-induced behaviors such as startle responses, obstacle navigation, compression–expansion events, and regrouping. Multi-object tracking will be used to estimate the position and velocity of each fish over time.

For a school of N fish, the individual state at time t is defined as:

$$\mathbf{x}_i(t) = \begin{bmatrix} p_{i,x}(t) \\ p_{i,y}(t) \\ v_{i,x}(t) \\ v_{i,y}(t) \end{bmatrix}, \qquad \mathbf{X}(t) = \{\mathbf{x}_1(t), \mathbf{x}_2(t), \ldots, \mathbf{x}_N(t)\} \tag{1}$$

This reconstructed state is not treated as the final object of analysis. Instead, it serves as the basis

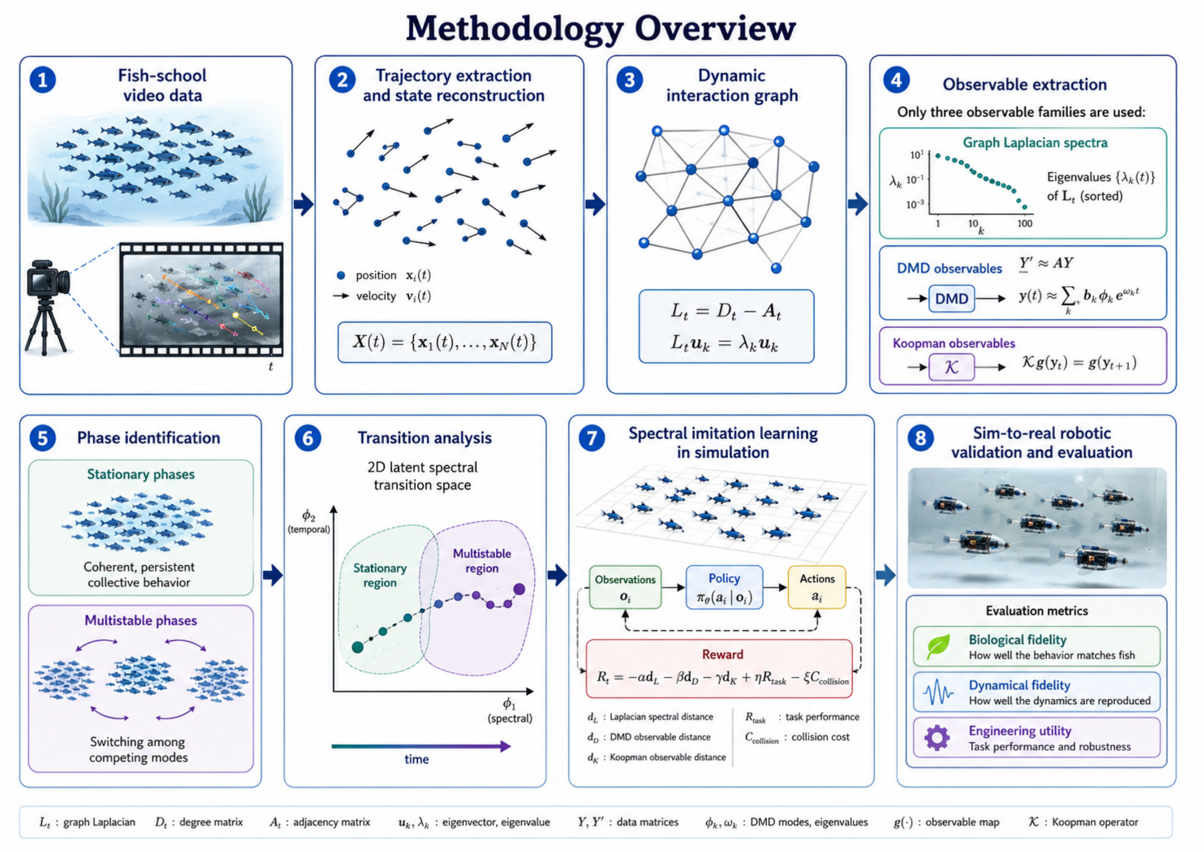

Figure 1: Methodology overview for the proposed spectral transition framework. Fish-school video data are transformed into trajectories, dynamic interaction graphs, and three observable families – graph Laplacian spectra, DMD observables, and Koopman observables – which are then used to identify stationary and multistable phases, analyze transition pathways, train simulated swarm policies, and validate transfer to robotic swarms.

for constructing graph Laplacian, DMD, and Koopman observables.

5.2 Graph Laplacian Spectral Observables

At each time point, the school will be represented as a dynamic interaction graph:

$$G_t = (V_t, E_t) \tag{2}$$

where each fish is a node and edges encode potential interaction relationships. Edges may be defined using k-nearest neighbors, distance thresholds, visual-field constraints, or inferred local influence. A simple k-nearest-neighbor adjacency is:

$$A_{ij}(t) = \begin{cases} 1, & j \in \mathcal{N}_k(i,t) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

The corresponding graph Laplacian is:

$$L_t = D_t - A_t \tag{4}$$

where D_t is the degree matrix. The spectral decomposition is:

$$L_t \mathbf{u}_k(t) = \lambda_k(t) \mathbf{u}_k(t) \tag{5}$$

The Laplacian eigenvalues and eigenvectors will be used as observables of the school's interaction topology. In particular, the second-smallest eigenvalue, $\lambda_2(t)$, will be used as a measure of algebraic connectivity. Higher values of $\lambda_2(t)$ indicate stronger graph cohesion, while values approaching zero may indicate fragmentation or subgroup formation.

The graph spectral observable vector may be defined as:

$$\mathbf{s}_L(t) = \begin{bmatrix} \lambda_2(t) \\ \lambda_3(t) \\ \vdots \\ \lambda_L(t) \end{bmatrix} \tag{6}$$

This representation will help detect whether transitions between stationary and multistable phases are associated with changes in interaction topology, such as loss of cohesion, subgroup formation, or reconnection after fragmentation.

5.3 Dynamic Mode Decomposition Observables

Dynamic Mode Decomposition will be used to identify coherent temporal modes in the collective dynamics. DMD may be applied either to the reconstructed state sequence $\mathbf{X}(t)$, to graph spectral vectors $\mathbf{s}_L(t)$, or to a combined observable vector constructed only from the allowed observable families.

Given a sequence of observable vectors, $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_m$, the snapshot matrices are:

$$Y = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \\ \vdots \\ \mathbf{y}_{m-1} \end{bmatrix}, \qquad Y' = \begin{bmatrix} \mathbf{y}_2 \\ \mathbf{y}_3 \\ \mathbf{y}_4 \\ \vdots \\ \mathbf{y}_m \end{bmatrix} \tag{7}$$

DMD estimates a linear operator $\mathcal{O}$ such that:

$$\mathcal{O} \phi_k = \mu_k \phi_k \tag{8}$$

where ϕ_k is the k-th DMD mode and μ_k is its associated eigenvalue. The continuous-time growth or oscillation rate is:

$$\omega_k = \frac{\log \mu_k}{\Delta t} \tag{9}$$

The observable dynamics can then be approximated by:

$$\mathbf{y}(t) \approx \sum_{k=1}^{r} b_k(t) \phi_k e^{\omega_k t} \tag{10}$$

DMD observables will therefore include modal amplitudes, modal energies, eigenvalue magnitudes, and oscillation frequencies:

$$E_k = |b_k(t)|^2 \tag{11}$$

Stationary phases are expected to show stable dominant DMD modes, while multistable phases are expected to show mode competition, intermittent dominance shifts, or the emergence of multiple active modes.

5.4 Koopman Observable Space

Because fish-school dynamics are nonlinear, Koopman-inspired observables will be used to describe the evolution of collective behavior in a lifted space. The Koopman operator acts on observables rather than directly on the raw state:

$$\mathcal{K}g(\mathbf{y}_t) = g(\mathbf{y}_{t+1}) \tag{12}$$

where g is an observable function and $\mathbf{y}_t$ is the state representation used for the analysis.

In this project, the Koopman observable vector will be constructed from the allowed observable families: graph Laplacian spectra and DMD-derived modal quantities. For example:

$$\mathbf{z}(t) = g(\mathbf{y}(t)) = \begin{bmatrix} \mathbf{s}_L(t) \\ \mathbf{E}_{DMD}(t) \\ \mathbf{s}_L(t) \otimes \mathbf{E}_{DMD}(t) \\ \mathbf{s}_L^2(t) \\ \mathbf{E}_{DMD}^2(t) \end{bmatrix} \tag{13}$$

The finite-dimensional Koopman approximation is:

$$\mathbf{z}(t+1) \approx K\mathbf{z}(t) \tag{14}$$

This lifted representation will be used to expose nonlinear relationships between topology, modal dynamics, and phase transitions. The goal is to obtain a coordinate system in which stationary regions, multistable regions, and transition pathways become more clearly separable.

5.5 Identification of Stationary and Multistable Phases

Stationary and multistable phases will be identified using the temporal behavior of the graph Laplacian, DMD, and Koopman observables. A stationary phase will be defined as an interval in which the observable structure remains approximately stable:

$$\|\mathbf{z}(t+\Delta t) - \mathbf{z}(t)\| < \varepsilon \tag{15}$$

A multistable phase will be defined as an interval in which several modes or spectral configurations coexist and compete. This will be quantified using normalized modal energy:

$$\tilde{E}_k(t) = \frac{E_k(t)}{\sum_j E_j(t)} \tag{16}$$

and modal entropy:

$$H(t) = -\sum_k \tilde{E}_k(t) \log \tilde{E}_k(t) \tag{17}$$

Low entropy indicates dominance of a small number of modes, while higher entropy suggests mode competition and possible multistability. In parallel, changes in the Laplacian spectrum will indicate whether multistability is associated with topological reorganization:

$$\Delta\lambda_k(t) = \lambda_k(t + \Delta t) - \lambda_k(t) \tag{18}$$

Thus, multistability will be characterized jointly as modal competition in DMD/Koopman space and topological reconfiguration in graph spectral space.

5.6 Transition Detection and Phase-Space Mapping

Transitions will be treated as the central dynamical events of the project. A transition will be identified when the school moves from one stable spectral region to another, or when stationary dynamics destabilize into multistable switching. The latent transition coordinate will be defined as:

$$\mathbf{z}_t = \Phi\left(\mathbf{s}_L(t), \mathbf{E}_{DMD}(t), \mathbf{z}_{\text{koopman}}(t)\right) \tag{19}$$

The resulting sequence, $\mathbf{z}_1, \mathbf{z}_2 \ldots \mathbf{z}_T$, will describe the school's trajectory through a spectral transition landscape.

Candidate transition markers will include:

$$\frac{d\lambda_2(t)}{dt}, \; \frac{dH(t)}{dt}, \; \frac{dE_k(t)}{dt}, \; \frac{d\mathbf{z}(t)}{dt} \tag{20}$$

These quantities measure changes in graph cohesion, modal entropy, DMD modal energy, and Koopman-state motion. A transition from phase q_i to phase q_j may be modeled as:

$$P_{ij} = \Pr(q_{t+1} = q_j \mid q_t = q_i) \tag{21}$$

This will allow the project to estimate which transitions are common, which are rare, and whether multistable regions act as intermediates between stationary phases.

5.7 Spectral Imitation Learning in Simulation

The learned observable space will be used as the objective for multi-agent reinforcement learning. Artificial agents will be trained to reproduce the graph spectral, DMD, and Koopman transition structure observed in fish schools.

Each agent follows a decentralized policy:

$$\mathbf{a}_i(t) \sim \pi_\theta(\mathbf{a}_i | \mathbf{o}_i(t)) \tag{22}$$

where $\mathbf{o}_i(t)$ is the local observation of the agent.

The reward function will compare biological and simulated swarms only through the selected observable families:

$$R_t = -\alpha\|\mathbf{s}_L^{\mathrm{bio}}(t) - \mathbf{s}_L^{\mathrm{sim}}(t)\|_2 - \beta\|\mathbf{E}_{DMD}^{\mathrm{bio}}(t) - \mathbf{E}_{DMD}^{\mathrm{sim}}(t)\|_2 - \gamma\|\mathbf{z}_K^{\mathrm{bio}}(t) - \mathbf{z}_K^{\mathrm{sim}}(t)\|_2 + \eta R_{\mathrm{task}} - \xi C_{\mathrm{collision}} \quad (23)$$

The policy will therefore be rewarded for reproducing not only fish-like collective organization, but also the observed transition geometry between stationary and multistable phases.

5.8 Robotic Transfer and Validation

After simulation, selected policies will be transferred to a physical robotic swarm platform. The physical swarm will be evaluated using the same observable families: graph Laplacian spectra, DMD modal structure, and Koopman-state trajectories. This ensures that biological, simulated, and robotic systems are compared in a shared dynamical language.

Sim-to-real degradation will be measured as:

$$D_{\mathrm{sim2real}} = \|\mathbf{s}_L^{\mathrm{bio}}(t) - \mathbf{s}_L^{\mathrm{sim}}(t)\|_2 + \|\mathbf{E}_{DMD}^{\mathrm{bio}}(t) - \mathbf{E}_{DMD}^{\mathrm{sim}}(t)\|_2 + \|\mathbf{z}_K^{\mathrm{bio}}(t) - \mathbf{z}_K^{\mathrm{sim}}(t)\|_2 \quad (24)$$

The main validation question is whether the dominant collective modes and transition pathways learned from fish schools persist when transferred to agents with different sensing, actuation, and communication constraints.

5.9 Evaluation

The framework will be evaluated according to three criteria: biological fidelity, dynamical fidelity, and engineering utility. Biological fidelity will measure whether the selected observables capture meaningful fish-school phase structure. Dynamical fidelity will measure whether stationary phases, multistable phases, dwell times, and transition pathways are preserved. Engineering utility will measure whether artificial swarms can perform tasks while maintaining biologically plausible transition dynamics.

Core metrics will include DMD reconstruction error:

$$\text{Reconstruction error} = \frac{\|Y - \hat{Y}\|_F}{\|Y\|_F} \quad (25)$$

graph spectral fidelity:

$$F_L = 1 - \|\mathbf{s}_L^{\mathrm{bio}}(t) - \mathbf{s}_L^{\mathrm{sim}}(t)\|_2 \quad (26)$$

modal fidelity:

$$F_D = 1 - \|\mathbf{E}_{DMD}^{\mathrm{bio}}(t) - \mathbf{E}_{DMD}^{\mathrm{sim}}(t)\|_2 \quad (27)$$

and Koopman-state fidelity:

$$F_K = 1 - \|\mathbf{z}_K^{\mathrm{bio}}(t) - \mathbf{z}_K^{\mathrm{sim}}(t)\|_2 \quad (28)$$

A successful outcome will be a framework that uses only graph Laplacian spectra, DMD, and Koopman observables to reveal the transition structure of fish-school collective dynamics and transfer that structure into artificial swarms.

6 Work Plan and Timeline

The research will be organized as a staged research program progressing from biological data acquisition and observable construction to transition analysis, learning-based swarm control, and robotic validation. The work plan is designed so that each stage produces a concrete methodological output and, where possible, a publishable research unit. The central thread throughout the project is the analysis of transitions between stationary and multistable phases in fish-school collective dynamics using three observable families: graph Laplacian spectra, Dynamic Mode Decomposition, and Koopman-inspired observables.

6.1 Year 1: Data, Observables, and Baseline Dynamical Characterization

The first year will focus on establishing the empirical and computational foundation of the project. This will include a focused literature review on fish schooling, collective motion, multistability, phase transitions in active matter, spectral graph theory, DMD, Koopman analysis, and bio-inspired swarm control. In parallel, suitable fish-school video datasets will be collected or curated from existing sources. The data will be selected to include both relatively stable collective behavior and episodes containing spontaneous or induced transitions, such as schooling-to-milling, compression-to-expansion, fragmentation-to-regrouping, and transitions into multistable switching.

A trajectory extraction and preprocessing pipeline will then be developed. Multi-object tracking will be used to reconstruct fish positions and velocities over time, with quality control procedures for occlusion, identity swaps, missing tracks, and noise. The reconstructed trajectories will be used to construct dynamic interaction graphs and time-series representations suitable for DMD and Koopman analysis. By the end of the first year, the project should have a validated dataset, a reproducible preprocessing pipeline, and initial descriptive analyses of stationary and transitional collective behavior.

6.1.1 Milestones for Year 1

- Complete literature review and theoretical framing.
- Curate or collect fish-school video datasets.
- Develop trajectory extraction and preprocessing pipeline.
- Construct dynamic interaction graphs from fish trajectories.
- Implement initial DMD and Koopman observable extraction.
- Produce preliminary characterization of stationary and transitional episodes.

6.1.2 Expected outcome - Year 1

A first methodological paper on the construction of spectral observables for fish-school collective dynamics.

6.2 Year 2: Spectral Phase Identification and Transition Analysis

The second year will focus on identifying stationary and multistable phases using the three selected observable families. Graph Laplacian spectra will be used to quantify interaction topology, cohesion, fragmentation, and subgroup formation. DMD will be used to identify coherent temporal modes, modal amplitudes, growth or decay rates, and oscillatory structure. Koopman-inspired observables will be used to construct a lifted dynamical representation in which nonlinear collective transitions can be analyzed more systematically.

The main objective of this year will be to define a latent spectral transition space. In this space, stationary phases should appear as relatively stable regions, while multistable phases should appear as regions of mode competition, repeated switching, or intermittent movement between multiple attractor-like structures. Transition events will be analyzed through changes in modal energy, DMD eigenvalues, Koopman coordinates, graph connectivity, and spectral entropy. Particular attention will be given to identifying early-warning markers that precede visible transitions between collective regimes.

6.2.1 Milestones for Year 2

- Define spectral signatures for stationary phases.
- Quantify multistable phases through modal competition and spectral entropy.
- Construct a latent spectral transition space.
- Detect transition events and estimate transition pathways.
- Identify candidate early-warning markers of phase change.

6.2.2 Expected outcome - Year 2

A core scientific paper on spectral characterization of transitions between stationary and multistable phases in fish schools.

6.3 Year 3: Spectral Imitation Learning in Simulated Swarms

The third year will transfer the biological analysis into a control framework for artificial swarms. A simulated multi-agent environment will be developed in which agents have local sensing, bounded actuation, noisy dynamics, collision constraints, and task objectives. The goal will not be direct imitation of fish trajectories, but reproduction of the spectral transition structure observed in biological schools.

Multi-agent reinforcement learning will be used to train decentralized swarm policies. The reward function will compare biological and simulated swarms using graph Laplacian spectral distance, DMD modal-energy distance, and Koopman-state distance. Additional reward terms will enforce practical constraints such as collision avoidance, energy efficiency, smooth motion, and task performance. The learned policies will be evaluated according to their ability to reproduce stationary phases, generate multistable switching, and preserve biologically plausible transition pathways under changing conditions.

This year will also include robustness tests. Policies will be evaluated across group sizes, noise levels, perturbations, sensing ranges, and environmental configurations. Baselines will include Reynolds-style attraction–alignment–repulsion rules, task-only MARL, trajectory imitation, and policies trained without explicit spectral transition rewards. The key question will be whether spectral imitation produces more biologically faithful and dynamically flexible swarm behavior than conventional approaches.

6.3.1 Milestones for Year 3

- Develop a simulated multi-agent swarm environment.
- Define spectral imitation rewards using graph Laplacian, DMD, and Koopman observables.
- Train MARL policies to reproduce biological collective phases.
- Train policies to reproduce transition pathways between phases.
- Evaluate robustness across noise, perturbations, and group sizes.
- Compare against classical and learning-based baselines.

6.3.2 Expected outcome - Year 3

A paper on spectral imitation learning for bio-inspired swarm control, with emphasis on reproducing transition dynamics rather than static formations.

6.4 Year 4: Robotic Transfer, Validation, and Thesis Integration

The fourth year will focus on sim-to-real transfer and thesis integration. Selected policies from simulation will be adapted to a physical robotic swarm platform, such as Kilobots or another available tabletop swarm system. Because physical robots are constrained by limited sensing, noisy actuation, communication restrictions, and localization error, simulation training will include domain randomization and robustness constraints before transfer.

The physical swarm will be evaluated using the same observable families used throughout the project: graph Laplacian spectra, DMD modal structure, and Koopman-state trajectories. This shared representation will allow biological, simulated, and robotic systems to be compared in a common dynamical language. The main validation question will be whether dominant collective modes and transition patterns can persist across different embodiments.

The final stage will integrate the biological analysis, spectral transition framework, MARL control system, and robotic validation into a unified dissertation. The thesis will emphasize the broader contribution of the work: showing that collective behavior can be understood not only by identifying the states a group occupies, but by analyzing the structured pathways through which it moves between them.

6.4.1 Milestones for Year 4

- Adapt trained policies for robotic swarm implementation.
- Perform sim-to-real transfer experiments.
- Evaluate robotic behavior using spectral transition metrics.
- Quantify sim-to-real degradation.
- Complete thesis chapters and dissertation integration.
- Prepare final publications and defense materials.

6.4.2 Expected outcome - Year 4

A robotic validation paper and the completed PhD dissertation.

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